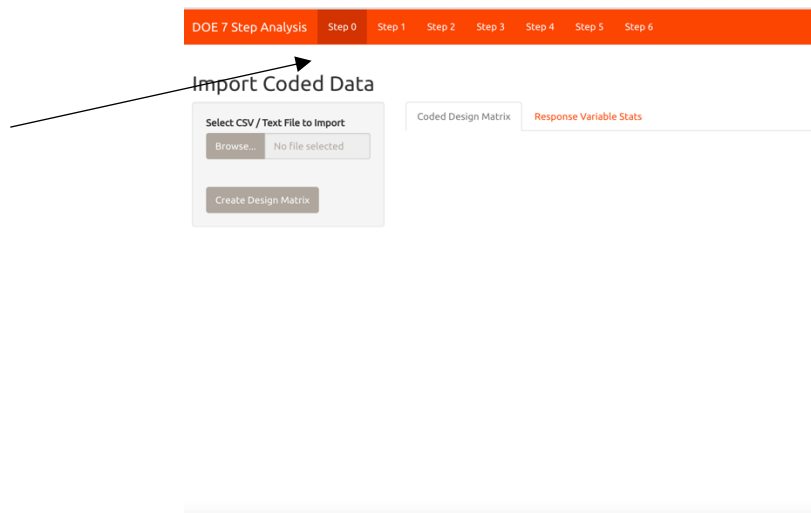


Table of Contents:

Step 0	2
Step 1	5
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Step 6	12

Step 0:

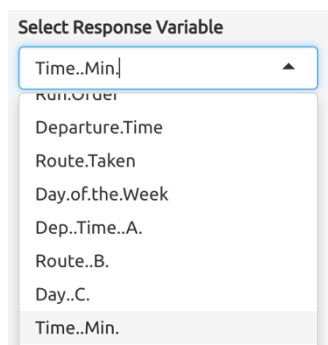
1. Navigate to the “Step 0” tab on the top panel



2. Click on the “browse” button and import a .csv or .txt file that contains your data



3. Select your response variable column from the drop down menu



4. Select your independent variables in the checkboxes (make sure you select the coded values)

Select Independent Variables

- ☐ Std.Order
- ☐ Run.Order
- ☐ Departure.Time
- ☐ Route.Taken
- ☐ Day.of.the.Week
- ☒ Dep..Time..A.
- ☒ Route..B.
- ☒ Day..C.
- ☐ Time..Min.

5. If you have blocks, select the blocking variable in the drop-down menu, otherwise, skip this step

Select Blocking Variable (Optional)

Not Selected

- Not Selected
- Std.Order
- Run.Order
- Departure.Time
- Route.Taken
- Day.of.the.Week
- Dep..Time..A.

6. Click on the “Create Design Matrix” button

Select CSV / Text File to Import

Browse... TimeToSchool.csv

Upload complete

Select Response Variable

Time..Min.

Select Independent Variables

- ☐ Std.Order
- ☐ Run.Order
- ☐ Departure.Time
- ☐ Route.Taken
- ☐ Day.of.the.Week
- ☒ Dep..Time..A.
- ☒ Route..B.
- ☒ Day..C.
- ☐ Time..Min.

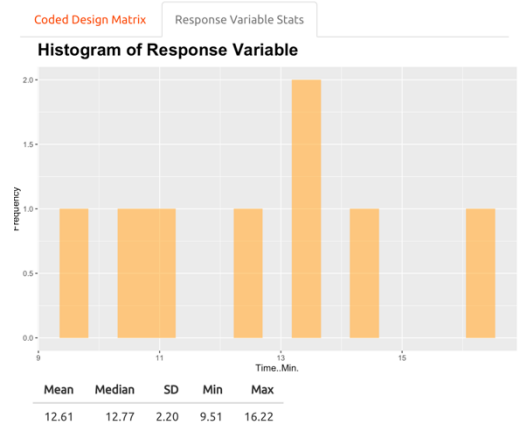
Select Blocking Variable (Optional)

Not Selected

Create Design Matrix

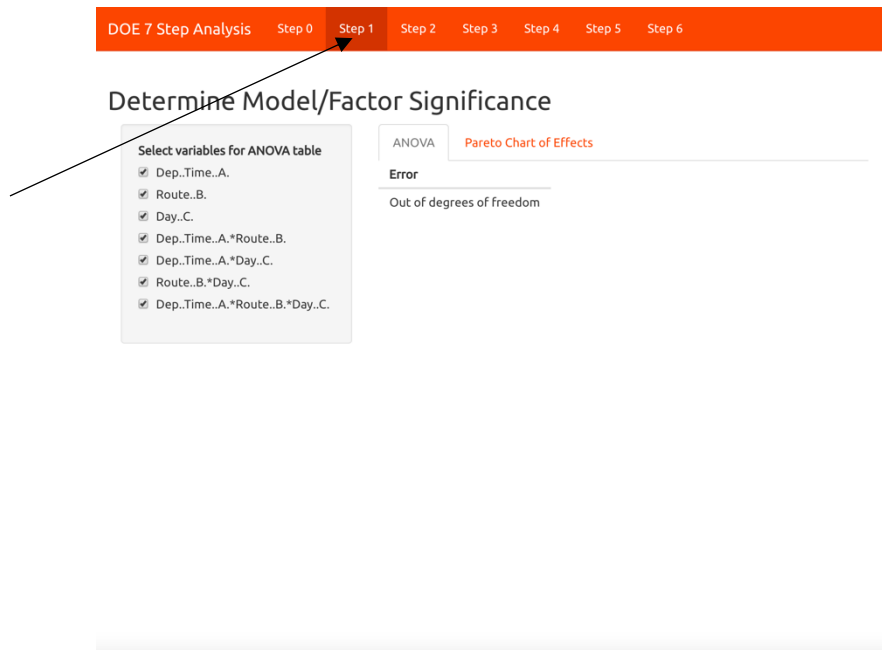
7. Click on the “Coded Design Matrix” tab or the “Response Variable Stats” tab to view you design matrix or information on the response variable

Coded Design Matrix		Response Variable Stats	
Dep..Time..A.	Route..B.	Day..C.	Time..Min.
-1	-1	-1	10.57
1	-1	-1	12.35
-1	1	-1	14.45
1	1	-1	16.22
-1	-1	1	9.51
1	-1	1	11.04
-1	1	1	13.19
1	1	1	13.55

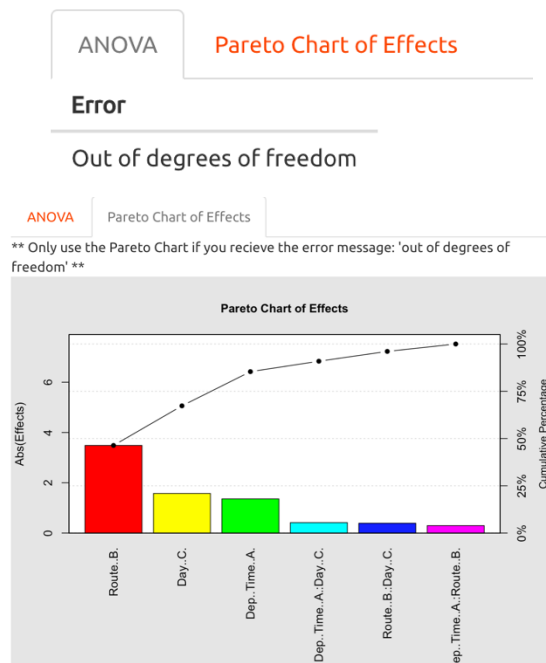


Step 1:

- Continue to “Step 1” on the top panel to view your ANOVA table



- If you receive this error, select the “Pareto Chart of Effects” tab to determine which interaction should be removed, otherwise continue



10. Perform you ANOVA analysis on the p-values by adding and removing factors and interactions using the checkboxes on the left

Select variables for ANOVA table

- ☒ Dep..Time..A.
- ☒ Route..B.
- ☒ Day..C.
- ☒ Dep..Time..A.*Route..B.
- ☒ Dep..Time..A.*Day..C.
- ☒ Route..B.*Day..C.
- ☐ Dep..Time..A.*Route..B.*Day..C.

Determine Model/Factor Significance

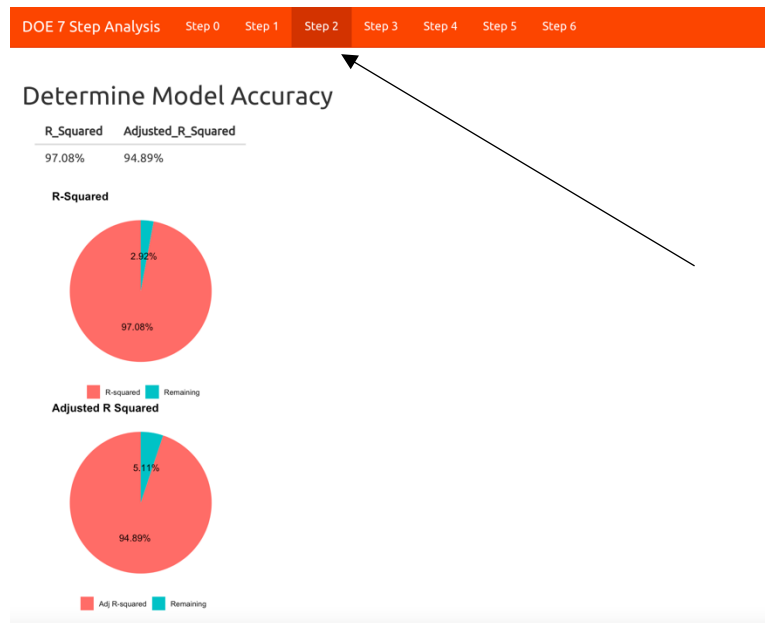
Select variables for ANOVA table

- ☒ Dep..Time..A.
- ☒ Route..B.
- ☒ Day..C.
- ☐ Dep..Time..A.*Route..B.
- ☐ Dep..Time..A.*Day..C.
- ☐ Route..B.*Day..C.
- ☐ Dep..Time..A.*Route..B.*Day..C.

ANOVA		Pareto Chart of Effects			
term	df	sumsq	statistic	p.value	
Dep..Time..A.	1.00	3.70	14.93	0.01808	
Route..B.	1.00	24.29	98.05	0.00058	
Day..C.	1.00	4.96	20.03	0.01103	
Residuals	4.00	0.99	NA	NA	
Total	7.00	33.94			

Step 2:

11. Continue on to the “Step 2” tab at the top to view your R-squared and adjusted R-squared values



Step 3:

12. Go to the “Step 3” tab to view your full model equation

DOE 7 Step Analysis Step 0 Step 1 Step 2 **Step 3** Step 4 Step 5 Step 6

Formulate Model Equation

term	estimate
(Intercept)	10.97
Dep..Time..A.1	1.36
Route..B.1	3.49
Day..C.1	-1.57

Time..Min. = 12.61 + 0.68 * Dep..Time..A. + 1.7425 * Route..B. + -0.7874999999999999 * Day..C.

Step 4:

13. Go to the “Step 4” tab to view your standard residuals and cook’s distance values

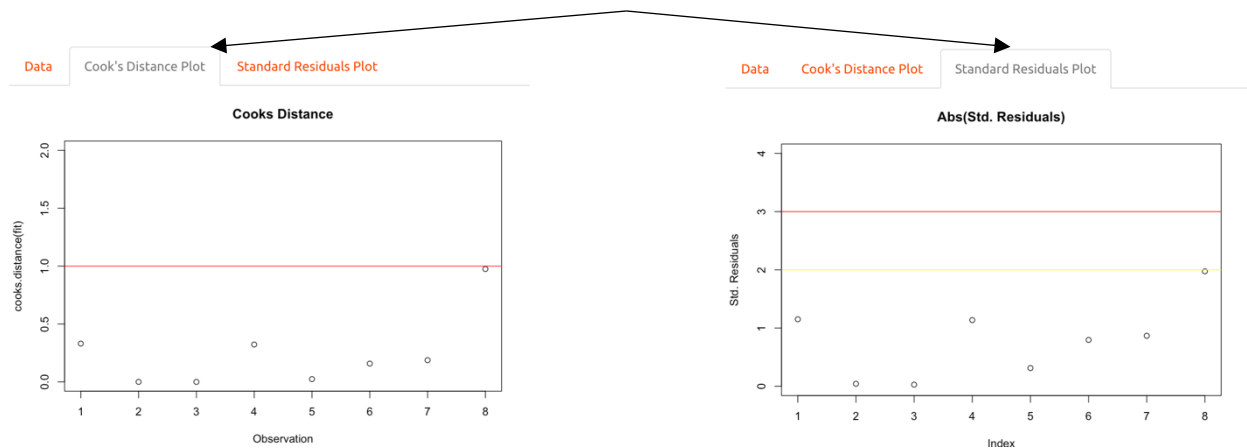
DOE 7 Step Analysis Step 0 Step 1 Step 2 Step 3 **Step 4** Step 5 Step 6

Analyze Residuals

Data **Cook's Distance Plot** Standard Residuals Plot

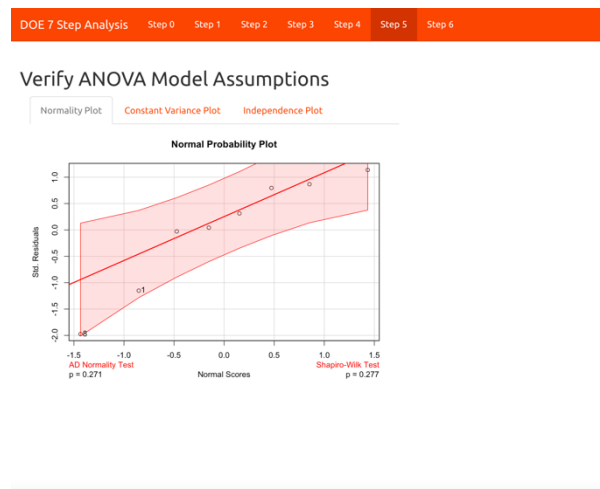
Dep_Time_A.	Route_B.	Day_C.	Time_Min.	std_residuals	cooks_distance
-1	-1	-1	10.57	1.151	0.331
1	-1	-1	12.35	0.043	0
-1	1	-1	14.45	0.028	0
1	1	-1	16.22	1.137	0.323
-1	-1	1	9.51	0.313	0.024
1	-1	1	11.04	0.796	0.158
-1	1	1	13.19	0.867	0.188
1	1	1	13.55	1.975	0.975

14. Select the “Cook’s Distance Plot” tab and the “Standard Residuals Plot” tab to visually confirm there are no outliers

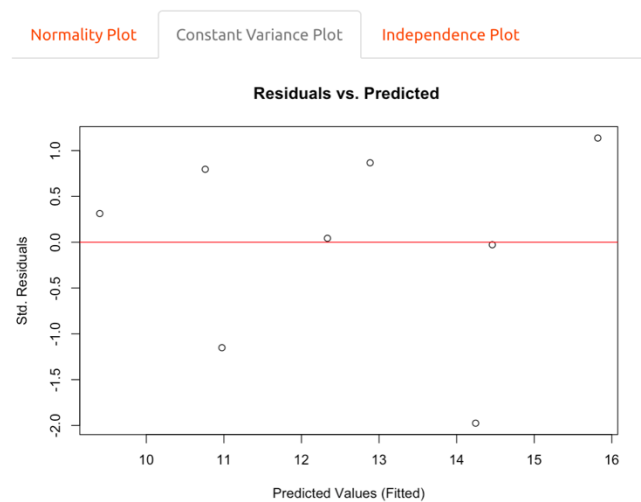


Step 5:

15. Go to the “Step 5” tab at the top
16. The page will open on the “Normality Plot” tab. Verify that your data is normal using the graph and the p-values



17. Select the “Constant Variance Plot” tab to view the variance plot and verify that your data has constant variance

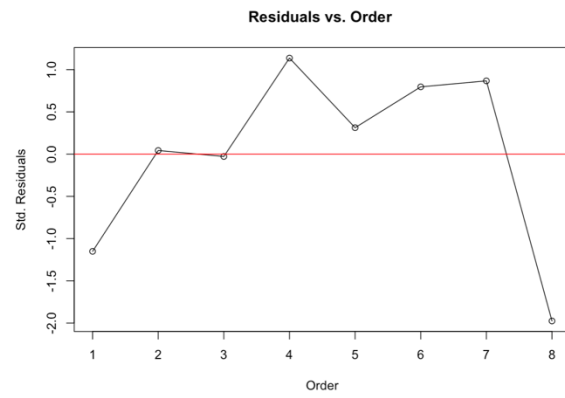


18. Select the “Independence Plot” tab to verify your data is independent

Normality Plot

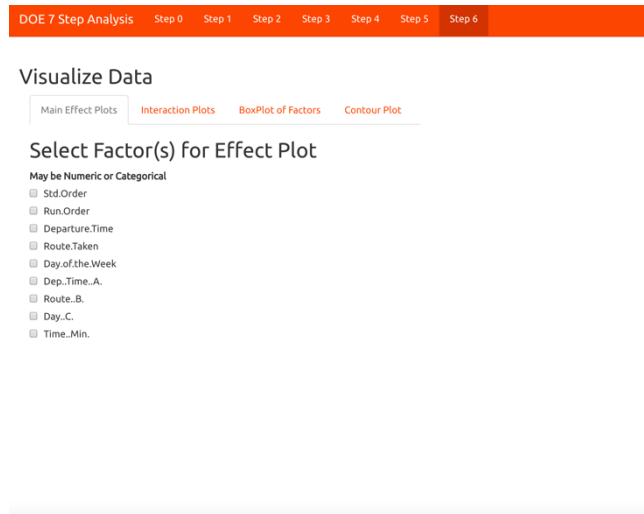
Constant Variance Plot

Independence Plot



Step 6:

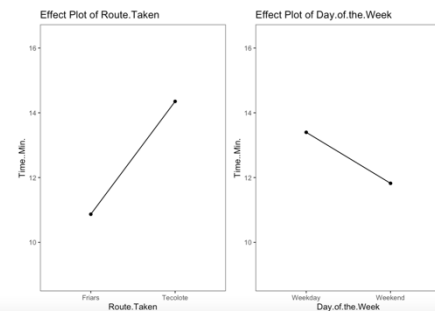
19. Select the “Step 6” tab at the top
20. Under the “Main Effect Plots” tab, select the factors you want to view the effect plot for using the checkboxes on the left



Select Factor(s) for Effect Plot

May be Numeric or Categorical

- ☐ Std.Order
- ☐ Run.Order
- ☐ Departure.Time
- ☒ Route.Taken
- ☒ Day.of.the.Week
- ☐ Dep..Time..A.
- ☐ Route..B.
- ☐ Day..C.
- ☐ Time..Min.

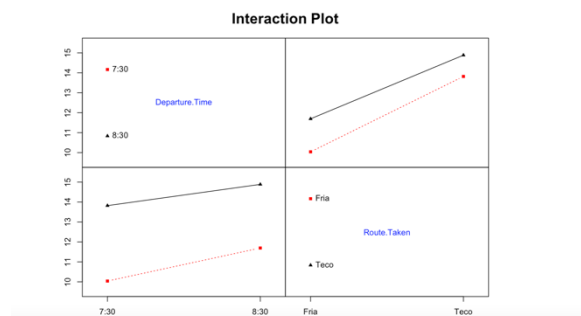


21. Under the “Interaction Plots” tab, select the factors you want to see the interaction plot for

Select Factors for Interaction Plot

Select 2-3 variables, Must be 2 levels, May be numeric or categorical

- ☐ Std.Order
- ☐ Run.Order
- ☒ Departure.Time
- ☒ Route.Taken
- ☐ Day.of.the.Week
- ☐ Dep..Time..A.
- ☐ Route..B.
- ☐ Day..C.
- ☐ Time..Min.

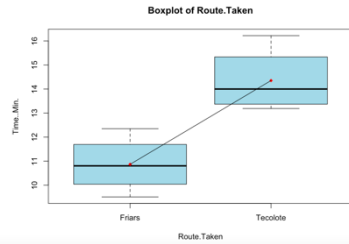


22. Under the “BoxPlot of Factors” tab, select the factor(s) you want to see in your boxplot

Select Factor(s) for Box Plot

May be Categorical or Numeric

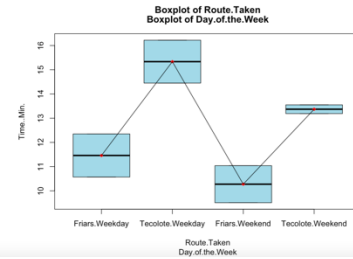
- ☐ Std.Order
- ☐ Run.Order
- ☐ Departure.Time
- ☒ Route.Taken
- ☐ Day.of.the.Week
- ☐ Dep.Time.A
- ☐ Route.B
- ☐ Day.C
- ☐ Time.Min.



Select Factor(s) for Box Plot

May be Categorical or Numeric

- ☐ Std.Order
- ☐ Run.Order
- ☐ Departure.Time
- ☒ Route.Taken
- ☒ Day.of.the.Week
- ☐ Dep.Time.A
- ☐ Route.B
- ☐ Day.C
- ☐ Time.Min.

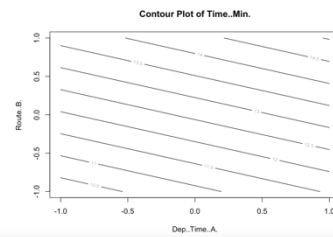


23. Under the “Contour Plot” tab, select the factors you want in your contour plot

Select Factors for Contour Plot

Only Select 2, Must be Numeric

- ☐ Std.Order
- ☐ Run.Order
- ☐ Departure.Time
- ☐ Route.Taken
- ☐ Day.of.the.Week
- ☒ Dep.Time.A
- ☒ Route.B
- ☐ Day.C
- ☐ Time.Min.



24. Formulate a conclusion for your data based on the graphs created and information found